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Automated Generation of Structure Datasets for Machine Learning Potentials and Alloys

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(Dated:)

We propose a strategy for generating unbiased and systematically extendable training data for machine learning interatomic potentials (MLIP) for multicomponent alloys, called *Automated Small SYmmetric Structure Training* or *ASSYST*. Based on exploring the full space of random crystal structures with space groups, it facilitates the construction of training sets for MLIPs in an automatic way without prior knowledge of the material in question. The advantages of this approach are that only cells consisting of few atoms (≈ 10) are needed for the DFT training set, and the size and completeness of the data set can be systematically controlled with very few parameters. We validate that potentials fitted this way can accurately describe a wide range of binary and ternary phases, random alloys, as well as point and extended defects, that have not been part of the training set. Finally, we estimate the binary phase diagrams with good experimental agreement. We demonstrate that the overall excellent performance is not a coincidence, but a consequence of the extensive sampling in phase space of ASSYST. Overall, this means that ASSYST will enable the largely autonomous generation of high-quality DFT reference data and MLIPs.

I. INTRODUCTION

Machine learning interatomic potentials (MLIP) are a topic of very active research, promising to bring the high accuracy of electronic structure methods to large space and long time scales previously reserved for empirical potentials. Recent years have seen impressive improvements in descriptor and regression techniques to the point where simple materials (without magnetism or charge effects) can be routinely described at meV/atom scale. [1–4] Very recently, potential frameworks have emerged that aim at describing the complete periodic table or significant portions of it, sometimes with impressive results. [5, 6] As more and improved potential models become available in the future, one fundamental problem remains—how to provide consistent and systematically expandable training data for a transferable potential. Where public databases of DFT results are reused, the included structures are very often stable or meta stable structures or derived from them. This means that phonons, defects, and transition states are rarely sampled.

As a consequence, MLIPs trained on such publicly available data are restricted in their training and have consequently difficulties extrapolating to non-equilibrium structures. This is a current bottleneck to the field of MLIP development. [7]. Multiple approaches exist to automatically sample the local environment of specific crystal structures under the umbrella of active learning (AL). [8–10] These are based on long sampling of molecular dynamics trajectories under a variety of boundary conditions and biases and iterative refitting of the poten-

tial models.

Our aim here is to show here that it is possible to generate training data and transferable potentials, *one shot*, i. e. without iterative refinement of the training set. This is desirable mainly for two reasons: Active Learning loops necessarily force a serialization of the training data and fitting workflow, and AL approaches can usually only guarantee local convergence. This means after a certain simulation finishes without new structures flagged by the AL method, there is no guarantee that the resulting potential accurately represents simulations at other temperatures, pressures or with other crystal structures. A method that cheaply generates a sufficiently diverse training set in one go instead would parallelize better and could offer better stability guarantees. In a previous paper [11] we described a systematic approach to generate training data for unary potentials and showed that the potential transferred well to different crystal structures and defects not explicitly included in the training set. In this work, we extend the method to an automated approach (*ASSYST*) applicable to arbitrary multi-component potentials. While doing so, we establish two major results: transferable and trustworthy potentials can be fitted with minimal human input and it can be done efficiently by training on small supercells consisting of only a few atoms. The human input is reduced to basic scalar convergence parameters, similarly to the energy cutoff in plane wave density functional theory.

Training on small cells has severe advantages in terms of required computational resources: Due to the cubic scaling of DFT and the fact that it only yields a single energy per structure, calculations of multiple smaller cells yield significantly more training data per compute time than larger cells. Because of this, this strategy has been investigated in other works in the literature as well. [12–

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14] A critical part of our verification will be to ensure that larger supercells are well represented by a training set consisting of only small cells.

In addition to our previous work[11], similar approaches have been advanced in the literature as well [15, 16] and in particular Yanxon et al. [17]. From the latter, we have used the crystal structure generation tool PYXTAL[18]. Liu and Mo [19] recently investigated the ability of Moment Tensor Potentials (MTP) to describe alloys when trained only on a few concentrations in the training set. Compared to these works, we extend the state of the art along a few directions. We explicitly and systematically define the structure generation workflow, identify convergable parameters and make it available as an open software package. We verify the resulting potentials in more depth and show their validity as general purpose potentials, rather than for crystal structure prediction. Crucially we can show that conventional AL would not extend training sets generated by *ASSYST*, i.e. they are complete in this sense. This places the automated generation of data by *ASSYST* on a solid basis and gives confidence for its application in the future.

The remainder of the article is structured as follows: we discuss *ASSYST* in depth in Section II A; briefly discuss the fitting of Moment Tensor Potentials (MTP) in Section II B. In Section II C we test the potential first on bulk properties. We then consider various point and extended defects in Sections II C 3 and II C 4. By benchmarking these not trained on structures against explicitly DFT computed ones we obtain detailed insight into the extrapolative power of this approach to upscale in size and concentration. Finally, we apply the MTP to compute and predict binary phase diagrams in Section II D. We discuss interpolation and extrapolation errors in Section II E. Necessary methods are reviewed in Section IV.

II. RESULTS

First we describe the details of *ASSYST*. Then we discuss the final training data and error metrics obtained from the fit. Next, we test the potential first on bulk and defect structures directly against DFT, before moving to the binary phase diagrams Mg/Al, Al/Ca, and Mg/Ca and compare them to experiment. Finally, we discuss whether the residual errors that we observe originate from interpolation or extrapolation errors.

A. Fundamentals of *ASSYST*

The initial structure generation is based on our earlier approach [11] and extends it to multicomponent systems. First, periodic structures with a given composition and space group symmetry are generated, then relaxed and finally randomly perturbed. Once all structures are obtained, highly converged DFT point calculations are

performed on all of them, with the parameters given in Section IV A.

1. *ASSYST*-Input

ASSYST requires very few external parameters to generate the training set. Also, the values we have identified change very little with the specific target system and can therefore be used as fixed (generic) parameters. The structure generation consists of three steps described by the following input parameters:

- the number of atoms, or the composition of possible starting configurations. We call this the *stoichiometry* in the following. This is the main convergence parameter. Higher values will sample phase space denser, but generate larger and computationally more costly training sets;
- how structures are relaxed specifically and whether samples are also taken along the relaxation trajectories;
- the amount of random perturbations to add to the final structure set.

In the following, we will explain each of the steps and their parameters in detail. Where they can be changed in our implementation, we will only give a variable name in the explanation here. In Section II A 7 we will introduce and discuss the numerical value we have chosen for this work.

2. Construction of Initial Structures

The main parameter of *ASSYST* is how many atoms are in the initial structures and what is their composition. Given a list of allowed occupations per element and structure, say 1–10 atoms, *ASSYST* first generates a list of all possible stoichiometries. For alloys we simply try all possible combinations, optionally limiting the total number of atoms per cell to n_{total} . Then for each of these stoichiometries, e.g. 2 Mg, 4 Al and 1 Ca, n_{SPG} random crystals are generated for each space group, 1–230. Of course, this could explode the number of structures very quickly if one were to allow large supercells, but we have found that structures of 8–10 atoms are generally sufficient to obtain good potentials. Not all stoichiometries can be realized in all space groups and naturally the simpler the atomic ratio (concentration) the more stoichiometries exist to realize them. This results in some concentrations being sampled more than others and leaves gaps where no structures are possible within the supercell limit we have chosen. We will verify in Section II C 5 that MLIPs fitted with *ASSYST* perform well also in these gaps.

3. Relaxation of Structures

This initial structure set is then relaxed within DFT at low convergence parameters. As the plain wave energy cutoff we use the largest of the element specific VASP[20, 21] defaults and sample reciprocal space with a minimum spacing between k points of $0.5 \text{ \AA}^{-1}$ (KSPACING=0.5 in the VASP[20, 21] INCAR file). First, we relax each structure with respect to volume, keeping cell shape and scaled atomic positions fixed. Then, we fully relax the resulting structures, i.e. allowing cell shape, size, and atomic positions to vary. The relaxations are performed as separate VASP[20, 21] runs to provide additional training points along the relaxation path and to avoid undue influence of Pulay stresses on the geometries. The final structures of both relaxation runs are collected and added to the training set separately. However, evenly spaced samples along the relaxation trajectory can also be taken. As mentioned above, the energies and other observables from these runs are discarded. Only the atomic structures on the relaxation path are used to recalculate energies, forces and stresses for the final training set with high convergence parameters.

The computational expense of this procedure is very modest compared to the final point calculations, allowing us to generate training structures that explore the potential energy surface without human input. Of course, multiple structures can relax to the same minimum (or close to each other), but we make no attempt to filter for such “redundant” structures. Intuitively, the wider the basin of attraction of a certain minimum energy structure, the more often a structure will fall into it, but also the more relevant it is for the material system. As such the redundant structures can help to sample the relevance of such basins automatically.

4. Adding Random Perturbations

Starting from the final set of fully relaxed structures we apply random perturbations on the cell shape, size, and atomic positions to sample the environment of minima on the potential energy surface thoroughly. Here again, the redundant structures increase the sampling of the relevant minima without resorting to human intuition or expert knowledge.

For every relaxed configuration, we generate n_{rattle} new structures with their positions changed by normally distributed noise with standard deviation σ_{rattle} and a small uniformly random strain matrix up to ϵ_{rattle} equal to 0.05 for diagonal (strain) entries and 0.005 for off-diagonal (shear) entries. This coupling between random displacements and small strains ensures that the fitted potentials do not contain unexpected holes that could be encountered e.g. when running NpT molecular dynamics simulations near the equilibrium structures.

Then again for every relaxed configuration we generate n_{stretch} new structures with their cells changed by ran-

dom matrices. Materials tend to be much stiffer in the shear directions than in the diagonal strain directions so sampling the diagonal strain is more important. Therefore we pick a “mostly diagonal” matrix in 70 % of the cases with diagonal entries uniformly distributed up to ϵ_{hydro} and off-diagonal entries up to 0.05 or in the other 30 % a “mostly off-diagonal” matrix with diagonal entries up to 0.05 and off-diagonal entries up to ϵ_{shear} .

5. Combining and Optional Filtering

The final training set is the combination of the initial random crystals, relaxation results, and perturbed structures. Because the structures generated especially during the initial phase can be significantly out of equilibrium it is important to filter them. Both PYXTAL[18] and RANDSPG[22] can sometimes generate unit cells of very high aspect ratio, i.e. the longest basis vector divided by the shortest one. We discard here any cell where this ratio exceeds an empirically found threshold of 6. We also make sure that the PAW spheres of the atoms do not overlap in any of our structures by checking the first nearest neighbor distances. This also helps the MTP fit, as otherwise the potentials have to learn the very steep but (usually) physically not very interesting repulsive interactions at close distances, which we have found can negatively affect their performance on other properties. Afterwards we may discard structures with very high energies or forces as well.

6. Intuitive understanding

One may ask how such a training set of very small supercells can possibly describe larger structures of practical relevance. We will show explicitly and quantitatively in Section II E that all defects and phases we consider in this work are not extrapolative, i.e. they are in a strong sense included in ASSYST generated training sets. Here we want to build up some intuition for this result. One can get a simple mental picture by realizing that the different radial functions used in MLIPs essentially slice the atomic neighborhood into different “generalized” shells. In the case of MTPs, this is done by the functions f_{μ} , see Eq. (2), but other potential types have similar functions. When predicting a new structure, descriptors made up of all radial functions are combined (linearly in the case of MTP) and thus can accurately “extrapolate” as long as each “slice” seen by the radial functions is represented in the training set.

We illustrate this here for the case of an interstitial in 2D in Fig. 1. In this figure, we plot the atomic density for three sample structures, one per row. The leftmost panel shows the full structure, while the three panels on the right are densities after convolving with Gaussian radial functions, f_i in the figure, centered on the colored circles shown in the left panels. Suppose we had two structures

in our training set, a bcc-like structure and a more complicated but still ordered tetragonal-like structure. We see that our three radial functions pick up different parts of the total atomic density. When our potential now tries to predict an interstitial (bottom row), we find that this (local) atomic density is actually very well approximated by a combination of f_0 from the bcc structure and f_1, f_2 from the tetragonal structure. This works for MTPs because the final potential is a linear combination of each descriptor function and assumes that the training set is sufficiently diverse to sample the descriptor space well, which is exactly what our method tries to do. The three columns on the right can also be thought of as graphical representations of an abstract feature matrix used to fit the potential. Since a MLIP based on atomistic density features then combines them again for the final prediction, we understand that a potential trained only on the first two rows (labeled “Training Set”) can accurately predict the last row (labeled “Prediction”). Of course, this is a toy illustration, and reality is significantly complicated by the fact that a point defect is not described by just one central atom, and radial functions are in general not simple Gaussians. But the above discussion makes it plausible that an isolated point defect can be approximated from data obtained only on periodic structures. In fact, since realistic radial functions may be negative and more spread out, real MLIPs would have more possibilities and could freely linearly combine atomistic densities seen during training.

Another, more abstract way to think about this is in terms of sampling the atomic density. A large number of MLIP descriptors, moment tensors among them, can be expressed as functions of a local atomic density

$$\rho(r) = \sum_i \delta(r - r_i),$$

where i enumerates atoms in an atomic structure with positions r_i . [23] A perfect training set would sample the space of all ρ evenly and densely. We can get some intuition for this sampling by considering instead the Fourier transform of the atomic density, $\tilde{\rho}(k) = \mathcal{F}[\rho]$. While this is not how MTPs are technically implemented, it provides the insight that the outer cutoff radius, R_c , of local potentials and the minimal distance enforced by Pauli repulsion between atoms, say $R_i < R_c$, essentially act as a band filter. Any atom centered local potential can only make use of information from $\tilde{\rho}$ if $|k| \in \left[\frac{1}{R_c}, \frac{1}{R_i}\right]$ and only this region needs to be sampled by a training set. If we exclude long-range effects such as unscreened Coulomb interactions, the “near-sightedness” of quantum mechanics [24] ensures that features at the short wavelength limit will be most important, i.e. represented by a small unit cell. Additionally, the “slicing” effect discussed in the previous paragraph, means that each descriptor function only needs to learn a small part of the k -dependency of $\tilde{\rho}$.

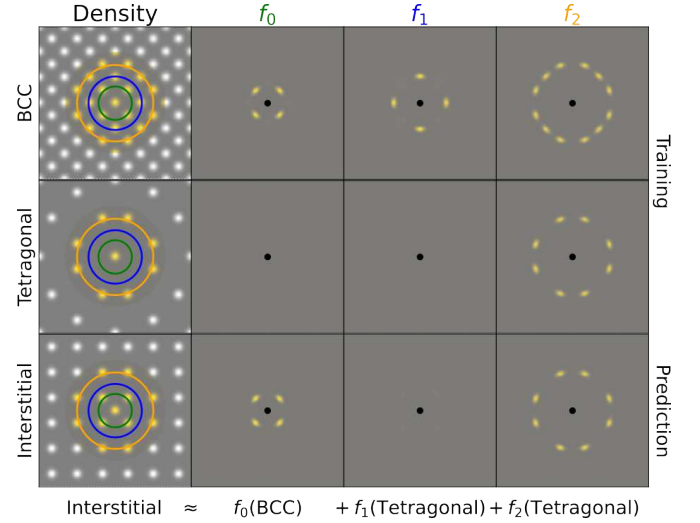


FIG. 1. **Cartoon atomistic density of three example 2D structures and interstitial** Shown are the atomistic densities (left column) and slices thereof (right columns) at different radii (colored rings in the left column) to illustrate how an MLIP based on three different radial descriptors can approximate an unseen point defect (here interstitial). The color scheme for the density is arbitrary, where gray indicates no atoms, yellow is used for atoms inside the cutoff of the potential descriptors, and white for atoms outside the cutoff. Black dots indicate the central atom in the sections in the right columns. The top row is for a “BCC-like” square centered structure, the middle row is for a more general and less dense “tetragonal-like” structure, and the last row is for a single interstitial in a primitive square lattice.

7. Parameters and Structures for this Work

The full list of hyperparameters and steps is given in Section IV D. After generating the structure set, we filter out all the data with small interatomic distances and high forces. Structures with pair distances smaller than the cutoff radii, r_i , per element are discarded, $d_{ij} < r_i + r_j$. We choose values for the radii close to the PAW sphere radii in the VASP [20, 21] pseudopotentials as $r_{\text{Mg}} = 0.9 \text{ \AA}$, $r_{\text{Al}} = 0.95 \text{ \AA}$ and $r_{\text{Ca}} = 1.15 \text{ \AA}$. To quantify “high forces” in this context we calculate the dimer curves of all combinations of Mg, Al and Ca and take the force acting on the dimer when the PAW spheres start to overlap. In our system these forces are of the order of $5 \text{ eV/\AA}$ to $10 \text{ eV/\AA}$ and so we filter all structures with forces higher than $5 \text{ eV/\AA}$.

In total, we obtain 116,690 structures and 655,766 atomic environments this way. The distribution of the training points in the ternary concentration space is shown in Fig. 2. Well visible is the discrete sampling that results from the small cells and their limited stoichiometry that we use for training. Changing the total number of atoms (n_{total} in the discussion above) it would be possible to arrive at denser or sparser training sets. This also controls the total number of structures

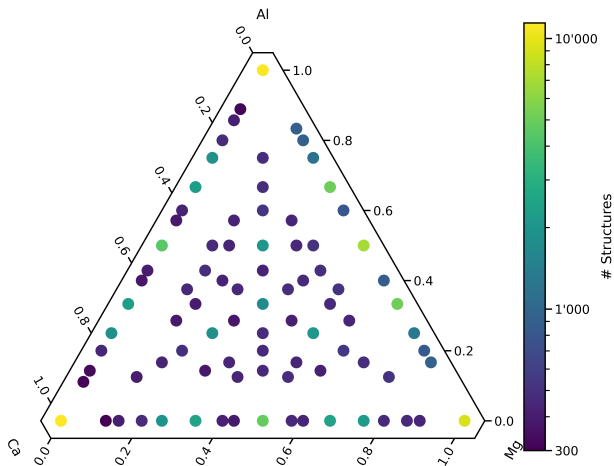


FIG. 2. **Stoichiometries present in Training Set** The distribution of stoichiometries in the training set, where the number of structures for each stoichiometry is shown on a logarithmic color scale. Each unary is represented by about 10,000 structures (9,109 for Mg, 11,244 for Al and 11,411 for Ca), the binaries by about 20,000 structures (24,837 for Mg/Al, 16,557 for Al/Ca, 18,161 for Mg/Ca) and the ternaries by 25,371 structures.

ASSYST generates.

In Fig. 3 we plot the full training set as a histogram over the atomic energies and volumes. The energies span a wide range of just over 10 eV/atom. In the histogram, we observe that the majority of points fall within $16 \text{ \AA}^3/\text{atom}$ to $42 \text{ \AA}^3/\text{atom}$ and -3.7 eV/atom to -1.5 eV/atom , corresponding to the bulk volumes and cohesive energies in this ternary system, cf. in Supplementary Note 1 and Table 1, though substantially more unfavorable configurations are covered as well. It is somewhat remarkable that *ASSYST* finds and covers this relevant range of structures well, but it may be understood in the sense that it is essentially a basic version of AB-INITIO RANDOM STRUCTURE SEARCH (AIRSS) [25]. As we will see below in Section II C 2 this property of the training set allows potentials to transfer exceptionally well to a variety of crystal structures.

We developed this procedure iteratively from pure Mg over binary Mg/Al to the full ternary system and as such there are some variations in the hyperparameters we ultimately used. For the pure Mg structures, we used the training set already published in Poul et al. [11] using RANDSPG[22]. For the Mg/Al structures we used a very similar procedure using RANDSPG[22] as well. The other systems follow the general method described in this work. The structures for pure Al were already part of another publication [26], though we have recomputed them here with consistent DFT settings.

Given the cost of large DFT datasets, we did *not* repeat the training set generation with a fully consistent set of parameters. Nevertheless, we do not observe significant differences in the quality of the description of the

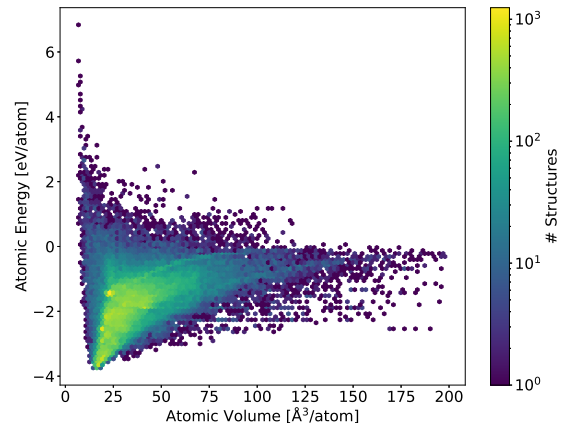


FIG. 3. **Energy-Volume Plot of Training Data** A histogram of the total training data used, binned on the atomic energy and volume. The x-axis is cut at $200 \text{ \AA}^3/\text{atom}$, only a few training points are beyond this range. The color scale is logarithmic. The concentration of training structures between $16 \text{ \AA}^3/\text{atom}$ to $42 \text{ \AA}^3/\text{atom}$ corresponds to the equilibrium range of the low energy structures in the ternary system Mg/Al/Ca.

MTP between elements, suggesting that there is some room for improvement in obtaining a minimally sufficient training set for alloys. We leave this exploration for future work.

B. Fitting Results and Training Errors

The MTP training errors are collected in Table I. We report there the Root Mean Squared Error (RMSE) and the Mean Absolute Error (MAE),

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_i^N (o_i - \hat{o}_i)^2}$$

$$\text{MAE} = \frac{1}{N} \sum_i^N |o_i - \hat{o}_i|,$$

where i runs over all observations. The hat indicates the DFT ground truth and o is the observable from which the error was calculated, such as energy per atom, force on an atom or the defect energy. The obtained energy MAE of 8.4 meV/atom indicates a good fit.

We do not report separate test errors as we do not split off from the training set a separate test set as is commonly done. Error metrics are strongly dependent on the underlying data distribution. Since our training data here is highly randomized and widely distributed, such errors are not immediately interpretable. High errors at extreme energies and volumes could skew these

TABLE I. **Training Errors** Root mean square (RMSE) and mean absolute errors (MAE) for energy and forces on the training set.

	Energy	Force
RMSE	13 meV/atom	65 meV/Å
MAE	8.4 meV/atom	33 meV/Å

TABLE II. **Test Errors on Bulk Properties** Mean absolute error (MAE) on basic properties (equilibrium energy, E_{eq} , volume V_{eq} , lattice parameters, a , bulk modulus, B , and its pressure derivative B') of 22 test structures.

	E_0	Ω_0	a	B	B'
MAE	7.6 meV/atom	$0.17 \text{ \AA}^3$	$0.02 \text{ \AA}$	1.3 GPa	0.43
relative	0.3 %	0.6 %	0.3 %	3 %	10 %

metrics to larger values, but might not impact applicability. Also, very similar well-described training points could hide deviations in less well-sampled phase space regions. Therefore we will construct a completely separate test set in Section II C and quote appropriate errors there.

C. Verification with respect to DFT

1. Structure Test Set

Our test set contains 2234 structures. They are described below and contain bulk properties, energy volume curves, point and planar defects, and sampling of solid solution phases. None of these structures entered explicitly into the fit. Moreover, these structures generally have larger unit cells or different stoichiometry than the training set structures, and so provide a relevant and challenging out-of-domain test set.

We start with a set of 22 structures from the MATERIALS PROJECT that are within 10 meV of the ternary convex hull. Some details are summarized in the Supplementary Table 1 and Fig. 4 shows a few selected structures.

2. Bulk Properties

All structures are relaxed separately with DFT and the MTP, before we obtain the equilibrium energy E_0 , the lattice parameter, a , and the atomic volume Ω_0 . Table II compiles the mean absolute error (MAE) on this test set. For these properties the MTP achieves relative errors below 1 %, indicating good transferability of the ASSYST-trained MTP to unseen structures.

Figure 5 shows the convex hull for the involved binary systems. Deviations from DFT are generally within the training error. However, this creates some issues for the

Mg/Al system, where the MTP overestimates the binding of the MgAl_2 compound. Due to the small enthalpy of mixing, 10 meV/atom to 20 meV/atom, in the range of training error itself, describing this correctly is a hard problem. Accurate predictions of the convex hull require not only low absolute errors, but also guarantees as to the relative ordering of the phases, something that is not accounted for by common regression algorithms. Methods to improve this are discussed in the literature [27, 28], but we believe them to be orthogonal to the problem of generating comprehensive training data that we study in this work.

We also check how well the potential reproduces elastic properties by applying volumetric strains to each of the test structures. For each of the 22 test structures we apply 40 hydrostatic strains between -70 % and 400 % volumetric strain. Where such large (negative) strains would cause the total volume to be less than $10 \text{ \AA}^3/\text{atom}$, we omitted them to avoid excessive k mesh requirements and overlapping PAW spheres. Figure 6 shows the E-V curves. From them we obtain the bulk modulus, B , and its derivative with respect to pressure, B' , for all 22 test structures according to

$$B = V \frac{\partial^2 E}{\partial^2 V} \quad \text{and} \quad B' = \frac{\partial B}{\partial P}$$

for both the potential and the DFT reference structures.

3. Point Defects

We obtain an extensive set of point defects by taking all structures from the test set introduced above, except the ternary structures, and systematically add vacancies, antisites and some interstitials to them. Symmetrically inequivalent lattice positions in each structure are obtained from SPGLIB[29] and then either removed (vacancies) or have their type changed (antisites). Tetrahedral and octahedral interstitials are identified by looking for positions in between lattice sites with a prescribed number of neighbors, high Voronoi volume and approximately regular connection polyhedra. [30] More complex, but perhaps physically more relevant interstitials such as dumbbells were not explicitly considered. This results in 249 different point defect structures.

Figure 7 (a) compiles the results of all defects considered. The overall MAE is around 136 meV. This error is comparable to the accuracy of other MLIPs in the literature, which have been fitted specifically for the description of point defects. [31]. The fact that the ASSYST-fitted MTP is able to describe these with comparable accuracy without being explicitly fitted to them supports the qualitative account of the interpolation behavior that we have sketched out in Fig. 1. It appears that once a training set is sufficiently diverse, MLIPs are able to successfully piece together different structural environments, even if the exact structure predicted is not present in any of the training structures.

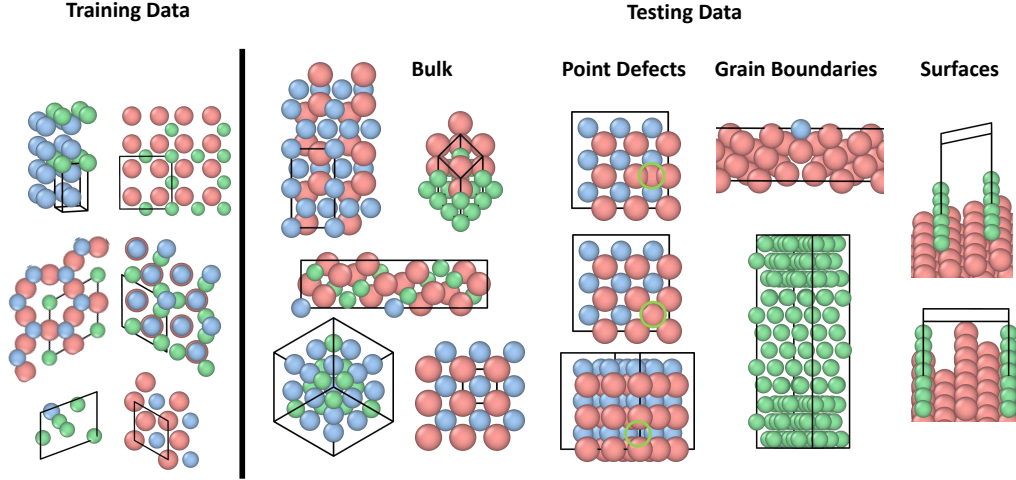


FIG. 4. **Example Structures present in Train and Test Data** Exemplary selection of structures present in the training and test sets, from left to right: training structures, bulk structures, point defects, planar defects. Some structures are shown as a supercell for clarity, where this is the case the original cell is drawn with black lines. Red circles highlight the exact location of the point defects.

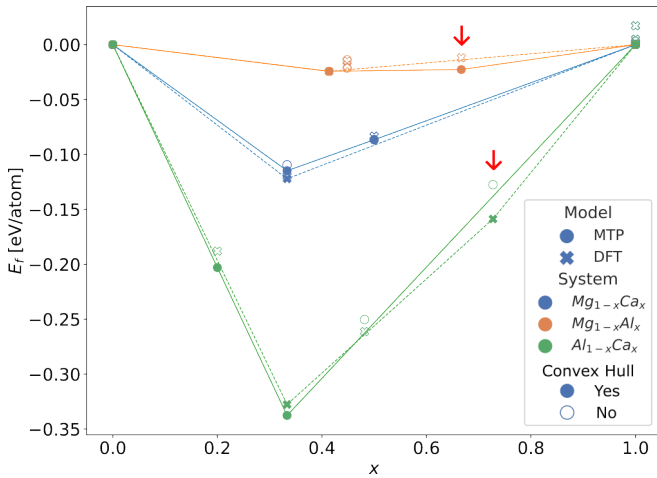


FIG. 5. **Convex Hull Diagrams predicted by MTP and DFT** Convex hull of the binary systems, Mg/Ca (blue), Mg/Al (orange) and Al/Ca (green). Solid lines are MTP predictions, dashed lines show the DFT verification results. Except for two deviations, Al_2Mg (mp-1094116) and Al_3Ca_8 (mp-1190736), marked with red arrows, the shape of the convex hull is well described.

4. Planar Defects

We now investigate more complicated planar defects and construct a set of surfaces and grain boundaries in the pure phases of all three elements, including segregated structures. Their construction is described in detail in the Supplementary Note 2.

A correlation plot of all 267 surfaces between MTP and DFT is shown in Fig. 7 (b). All energies are obtained

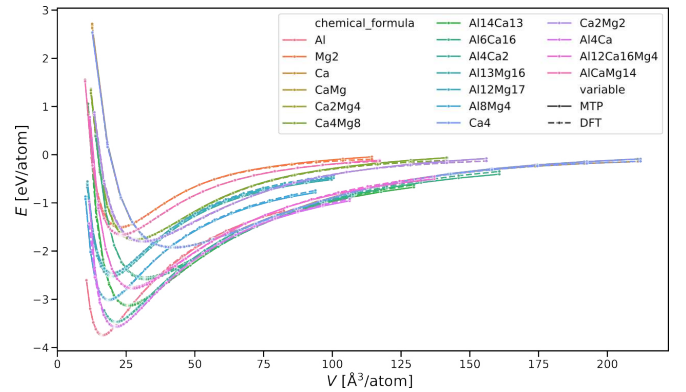


FIG. 6. **Verification of MTP predicted Energy-Volume Curves against DFT** Energy volume curves for all 22 test structures. Colored lines give the DFT reference (dashed) and the MTP predictions (solid). Dots show the volumes of the individual testing points. The MTP describes all phases well and over a wide range of volumes, even for those phases that are substantially larger than the small-size training structures.

from a single static calculation, i.e. without relaxation of any kind. MTP and DFT agree very well, especially considering that no defect or surface structures were explicitly included in the training. While surprising at first glance, we attribute it to some structures in the training set that delaminate during the volume relaxation steps of the training set construction.[11] These structures form quasi-2D crystals separated by vacuum and are apparently sufficient to provide enough information for the potential to extrapolate to the true surface structures.

Figure 7 (c) plots the error correlation of all 147 grain

boundaries. To avoid too close atoms we relaxed each grain boundary with the MTP before comparing the defect energies obtained by point calculations between DFT and MTP. The MTP achieves an MAE of $0.9 \text{ meV}/\text{\AA}^2$, indicating again that our training set covers a wide range of local environments that are sufficient to describe planar defects.

5. Concentration Interpolation

We now turn to the question of how well the MTP can interpolate between the coarse concentration grid of the training set and the finer concentrations available in larger structures. To this end, we prepare test sets based on the hcp, fcc, and bcc lattices. For all three lattices we prepare Special Quasirandom Structures (SQS) [32] with 32-54 atoms for the binary and 16-32 atoms for the ternary compositions and relax the volume of the structures with DFT and MTP independently. The initial volumes for the structures were set according to Vegard's law, starting from the volume of the stable elemental bulk phases. This results in 99 structures for each of the binaries and 324 structures for the ternary, or a total of 603 structures. We calculate the formation energy for every prototype and every concentration by

$$E_f(c_{\text{Mg}}, c_{\text{Al}}, c_{\text{Ca}}) = E(c_{\text{Mg}}, c_{\text{Al}}, c_{\text{Ca}}) - \sum_{X \in \{\text{Mg}, \text{Al}, \text{Ca}\}} c_X E_X$$

where $E_{\text{Mg}}, E_{\text{Al}}, E_{\text{Ca}}$ are the per atom energies of the stable prototypes for all three elements at their equilibrium volume. The obtained errors in E_f are shown in Fig. 8. They are generally within 5 meV/atom to 10 meV/atom , except for the Al-rich bcc structures, where errors can be up to 25 meV/atom . Since sampling in concentration space is uniform and since we detect no extrapolation for any SQS structure, c.f. Fig. 2 and Section II E, it appears that the potential energy surface of Al-rich bcc structures is harder to fit than for the other structures, though it is not particularly relevant for applications, since this phase is thermodynamically unstable at standard conditions.

D. Phase Diagrams

As a first practical application, we compute the temperature-composition binary phase diagrams of the Mg/Al/Ca system. Much effort could be expended here to obtain high-quality phase diagrams. However, for the purpose of potential validation, we will choose computationally efficient approximations that can be scaled over a large number of potentials during the testing phase and that can serve as a baseline for future research.

We will construct our phase diagrams using a set of structures obtained from the Materials Project [33] and calculate their atomic Gibbs free energy $G(T, p = 0)$ with CALPHY [34]. The chosen prototype phases are compiled in the Supplementary Note 3 and Table 2 and the

construction of the phase diagrams is described in Section IV G.

Non-stoichiometric behavior is modeled by including all possible antisite point defects. For computational efficiency, we consider only the excess enthalpy of these point defects without computing their temperature dependence. Our restriction to antisites is not general, but sufficient for the binary systems considered here. Al/Ca and Mg/Ca consist entirely of line phases, leaving only fcc-Al, hcp-Mg and β' - Mg_2Al_3 as phases with finite solubilities. Vacancy mediated off-stoichiometry is not possible for the terminal phases and has been ruled out for γ - $\text{Mg}_{17}\text{Al}_{12}$ by Shin and Wolverton [35]. A comparison between defect enthalpies and entropies in the same work shows that even at 1000 K the defect entropy is only 10 % of the defect-free energy for typical point defects in Mg/Al, suggesting to us that for a first order approximation of the phase diagram neglecting the entropies is justified. In the Supplementary Note 4 we estimate the effect of errors in the free energy on the transition temperatures in pure calcium. As expected, even small differences can lead to qualitative changes, so we leave the exact calculation of the phase diagrams of the MTP to future work.

1. Mg/Al

The relevant phases for this system are hcp Mg, fcc Al, β' - Mg_2Al_3 , γ - $\text{Mg}_{17}\text{Al}_{12}$ and ϵ - $\text{Mg}_{23}\text{Al}_{30}$. [36] Additionally, we included the MgAl_2 phase, which is experimentally metastable [37] but on the DFT convex hull according to the Materials Project [33]. Figure 9 shows the phase diagram calculated using the above approach with the MTP. Compared with recent *ab-initio* calculations, CALPHAD assessments and experiments [36–41] the qualitative agreement is very good, reproducing the melting temperatures and liquid-solidus lines well, within about 50 K to 100 K. β' - Mg_2Al_3 shows a finite solubility toward the aluminum side. While most assessments model this phase as a line compound, some experiments do report a similar solubility [40], but the data are generally rare. ϵ - $\text{Mg}_{23}\text{Al}_{30}$ is shifted to lower temperatures as a whole and the asymmetry of β' - Mg_2Al_3 is exaggerated. In assessments, this phase shows very low solubility towards the Mg-rich side, but maintains a constant phase boundary, whereas the MTP prediction shows a receding boundary towards the Al-rich side. Connected to this, the stability of the liquid is overestimated pushing the γ - $\text{Mg}_{17}\text{Al}_{12}$ + hcp-Mg $\rightarrow$ liquid eutectic point down and connecting the γ - $\text{Mg}_{17}\text{Al}_{12}$, β' - Mg_2Al_3 and the liquid in a peritectic rather than eutectic point. It seems likely that this is due to our simplified modeling of point defects, which may fail at high temperatures in this phase. The solubility is too low on the fcc Al side, probably because we assumed the defect-free energy of formation to be constant rather than decreasing at higher temperatures due to its excess entropy. However, all experimental phases are present

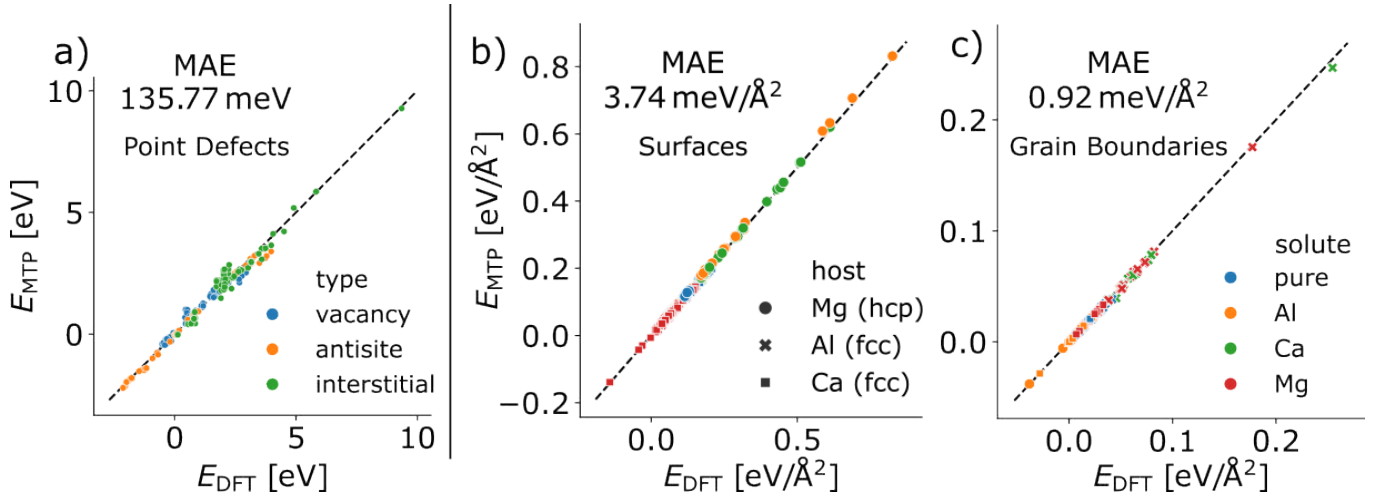


FIG. 7. **Correlation plots for the defect energies** a) Defect energy between DFT and MTP for 249 point defects, including vacancies, antisites and interstitials. Defect geometries are relaxed separately for DFT and MTP for this comparison. The MAE is 136 meV, with the largest contribution from interstitials due to their highly unfavorable nature in this material system. b) Unrelaxed cleavage energies for pristine and segregated surfaces of hcp Mg, fcc Al and fcc Ca with $h, k, l \leq 3$ between MTP and DFT. This data set includes flat, densely packed surfaces as well as corrugated and reconstructed surfaces as provided by PYMATGEN for a total of 264 structures. The agreement is remarkably good for all considered surfaces with an overall MAE of less than $4 \text{ meV}/\text{\AA}^2$. c) (MTP-)relaxed grain boundaries of hcp Mg, fcc Al and fcc Ca. The agreement is again remarkably good for all considered grain boundaries with an overall MAE of around $0.9 \text{ meV}/\text{\AA}^2$.

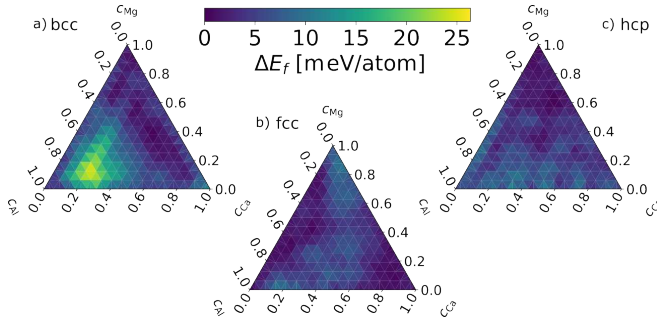


FIG. 8. **Accuracy of MTP for Solid Solutions** Errors in MTP predicted formation energy over the full ternary composition space for bcc (a), fcc (b) and hcp (c) structures are plotted in ternary diagrams.

and of the qualitatively expected shape. All three intermetallic phases have significantly larger unit cells than the training data, indicating that our small training cells are sufficiently diverse to sample a very large part of the phase space.

As a reference, we repeat all calculations with a Modified Embedded Atom Potential (MEAM) [42]. The computed phase diagram is shown in Fig. 9 (c). The solubilities of the terminals follow a similar trend as the ones for the MTP, suggesting that likely our simplified modeling limits the comparison to experiment here. Generally, the liquid is too stable and liquidus lines are about 100 K too low. Interestingly, the γ - $\text{Mg}_{17}\text{Al}_{12}$ is missing here, although it is reported to be on the convex hull at $T = 0 \text{ K}$. In addition, the Al_2Mg phase is strongly overstabilized,

displacing the β' - Mg_2Al_3 and ϵ - $\text{Mg}_{23}\text{Al}_{30}$ phases from the phase diagram. To partially remove these artifacts, we have constructed a metastable phase diagram where we have manually removed the overstabilized phase (see Supplementary Note 5 and Figure 2). This illustrates that even for the EAM or MEAM class of potentials, incomplete training data can easily lead to erroneous extrapolation. In contrast, the MTP *ASSYST*-fitted presented here does not suffer from such problems, even though the construction of the training set does not explicitly consider structures on the convex hull.

2. Mg/Ca and Al/Ca

Both the Mg/Ca and Al/Ca binary systems exhibit only line phases with essentially no compositional variation. The C14- Mg_2Ca , bcc-Ca+Mg, hcp-Mg+Ca, bcc/fcc-Ca+Al and C15- Al_2Ca phases show finite solubility in the latest CALPHAD evaluation [36], but less than 1 %. Here we model both systems fully in terms of line phases together with a regularly mixing melt. Details are given in Supplementary Note 3 and Table 2.

The predicted phase diagrams are shown in Figs. 10 and 11. Most notably, the MTP misses the AlCa and Al_3Ca_8 phases, but otherwise, the critical points and reaction temperatures are reproduced within approximately 100 K.

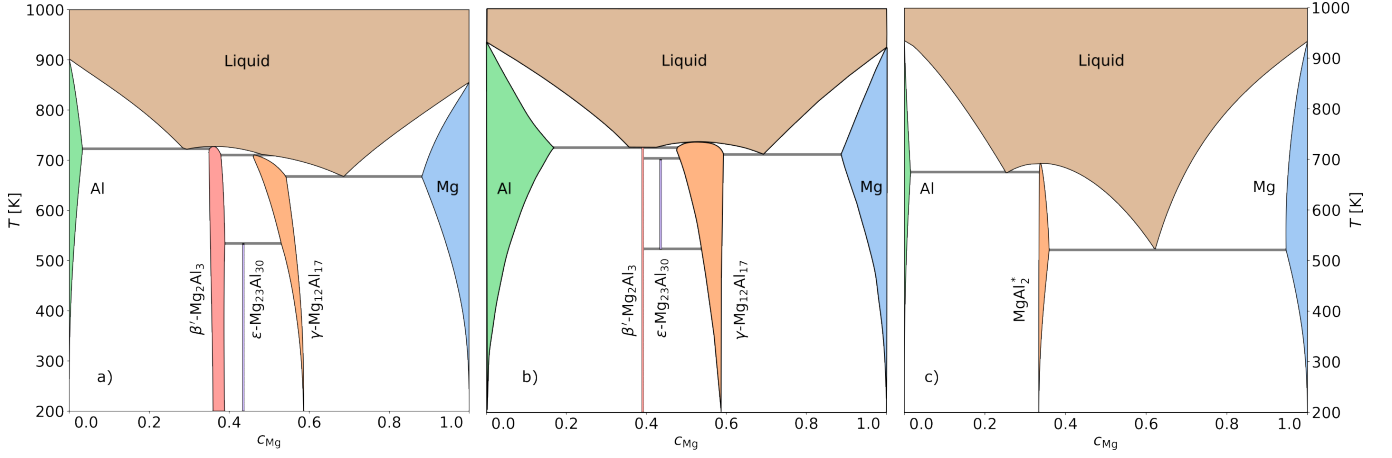


FIG. 9. **Mg/Al Phase Diagram** (a) The phase diagram of Mg/Al as predicted by the MTP. The structure prototypes used are shown in the Supplementary Table 2. Experimental phases are qualitatively correctly predicted despite approximate thermodynamic modeling. (b) CALPHAD assessment from [43]. Adapted and redrawn with permission from Elsevier. (c) The phase diagram of Mg/Al as predicted by a MEAM potential [42]. This potential overstabilizes the MgAl_2 phase and suppresses β' - Mg_2Al_3 and ϵ - $\text{Mg}_{23}\text{Al}_{30}$. The phase is normally metastable, indicated by a star. The γ - $\text{Mg}_{12}\text{Al}_{12}$ phase is underestimated by the potential at finite temperatures and does not appear naturally on the phase diagram. See Supplementary Figure 2 for a phase diagram where MgAl_2 was manually omitted.

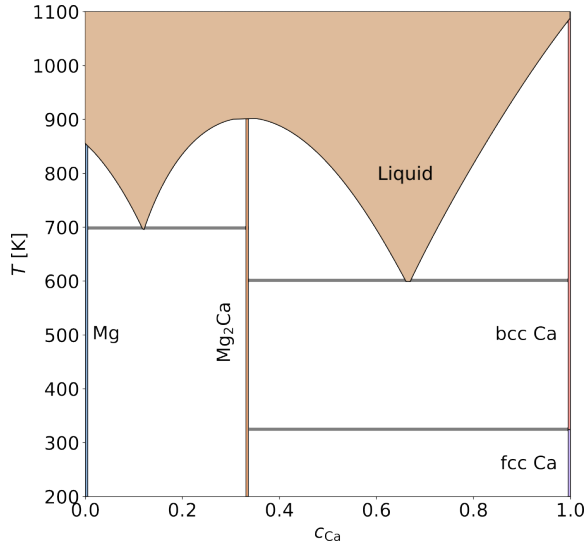


FIG. 10. **Mg/Ca Phase Diagram** The phase diagram of Mg/Ca as predicted by the MTP. The fcc to bcc transition in pure Ca is strongly underpredicted, but all other critical points are well described with respect to CALPHAD [36].

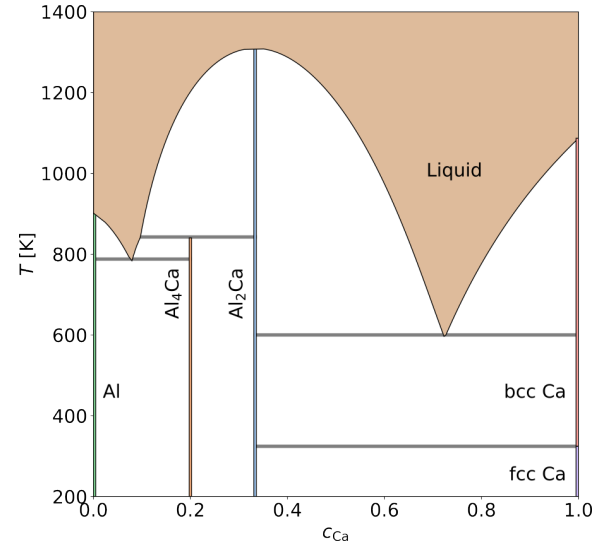


FIG. 11. **Al/Ca Phase Diagram** The phase diagram of Al/Ca as predicted by the MTP. The phases AlCa and Al_3Ca_8 are missing, but other features of the current CALPHAD predictions [36] are well represented.

E. Interpolating vs. Extrapolating

We wish to show that the *ASSYST* generated training set is complete in the sense that for a wide variety of applications the potentials remain stable and give accurate results without continuous adjustment. What we have presented so far implies that the MLIP guarantees reliable results over a wide range of conditions, but the question remains whether this is really a consequence of the training set or rather a peculiarity of the material system

we are working with. In other words, is the MLIP, e.g. in Fig. 7, extrapolating well or interpolating in between training configurations? The latter would imply that we can expect this procedure to generate training data to transfer to other material systems as well. The MLIP-2[44] package optionally reports the extrapolation grade of a structure with respect to a given training set. These grades are larger than one when a potential is extrapolating and smaller than one when interpolating. [45, 46] We collect first all structures used in the verification in

Section II C, a total of 2234 configurations, and calculate their extrapolation grade. Less than 5 % of the structures have an extrapolation grade larger than one, i.e. are in the extrapolative regime. Only 1 % exceed grades larger than two, a limit that is commonly quoted for reliable extrapolation. The maximum grade was 8.75 for some highly compressed Al structures. All cases of high extrapolation grades occur exclusively in the strongly repulsive limit. More details are shown in Section IV F. It is striking that none of the structures tested in Section II C 5 are extrapolative, which implies that *ASSYST* generated data sets cover the full space of concentrations despite their small structure sizes.

We also obtained all structures from Materials Project[33] containing only Mg, Al and Ca, an additional 150 structures. Here the maximum grade was 1.23, with only three structures having extrapolation grades larger than one. This shows that the training set is indeed complete, and the procedure for generating training structures can be expected to transfer well to other material systems.

Additionally as part of the free energy calculations for Section II D we ran MD for multiple hundred thousand timesteps for multiple phases over a wide range of temperatures without failure or instability or the need to refit the MLIP, even for liquids. This implies that the *ASSYST* based MLIP is robust, stable and can be used for a wide range of applications.

Any errors on structures that do not show extrapolation must consequently be interpolation errors. Since we do not observe significant extrapolations here, the errors e.g. in Figs. 5 and 8 must also be interpolation errors. The description of these quantities might be improved by adding corresponding structures manually or by running *ASSYST* with a larger n_{total} and/or increasing descriptor basis set.

III. DISCUSSION

In the literature, a number of machine learning potentials have been fitted to similar systems in particular Mg/Al, but often with a different ternary element.[47–49]. Embedded Atom Method (EAM) and Modified Embedded Atom Method (MEAM) exists [42]. We compared explicitly to this potential in Section II D. A shortcoming of it is that it overestimates the binding of the experimentally metastable compound MgAl_2 which displaces the other intermetallic compounds from the phase diagram.

The Neural Network Potentials (NNP) of the Behler-Parrinello type fitted by Marchand and Curtin [47], Jain et al. [48] are well constructed and thoroughly tested, yet rely on deep metallurgical knowledge to construct the training sets. Here we achieve comparable results on several quantities, with a fully automated training set, while also significantly extending the strain range at which the potentials stay stable.

The network potential of the deep neural net type

(DNP) fitted by Andolina et al. [49] was specifically constructed to study surface segregation based on a manually curated training set filled by ab-initio MD simulations performed by VASP[20, 21]. Using our approach and the MTP model, we achieve an improved description of the surface energies and several bulk qualities with a significantly smaller training set.

A. Conclusion

We presented an automated method for generating training data for machine learning potentials. Potentials fitted to this data showed very good stability and transferability. Remaining errors are mainly interpolation errors. In particular and worth stressing is that the MTP was also able to accurately describe structures that are larger than structures included in the training set or of different stoichiometry and that these structures did not trigger the built-in active learning algorithm of MTPs. In Section II E we provide a tentative explanation, real materials and their defects share a significant overlap of *local structure* and it is possible to sample this space because its size is bounded by the symmetry constraints of 3D periodic structures. While this still does not guarantee that a potential fitted to such a sampling would be homogeneously accurate across, it does shift the focus from extrapolation errors to interpolation errors.

We have demonstrated the practical applicability of the MTP by estimating the Mg/Al phase diagram in good qualitative agreement with experiments and previous theoretical studies, despite basic thermodynamic modeling. This also shows that the specific MTP trained here will allow in the future detailed studies of complicated phases, such as $\gamma\text{-Mg}_{17}\text{Al}_{12}$ and $\beta'\text{-Mg}_2\text{Al}_3$, which have rarely been investigated in the literature, because complicated structure and a lack of adequate potentials.

Our focus on small training structures will be particularly useful, if the field moves towards references beyond DFT, such as Coupled Cluster methods[50], or corrections to it such as Random Phase Approximation calculations[51]. Such approaches are starting to appear in the literature[52–55], usually using transfer learning. Such methods benefit greatly from training on small cells, since more accurate electronic structure methods scale even worse than DFT. Our work here clearly shows that such small cell data are sufficient to describe the behavior of much larger condensed matter systems.

We believe that diverse and accurate training data will be a crucial ingredient in the upcoming development of *ab initio* accurate, scalable machine learning potentials, regardless of which potential architecture or regression technique will prevail in the next few years. *ASSYST* automatically searches the phase space of the target material system and provides such training data.

IV. METHODS

A. DFT Data Generation

All training data is generated using VASP[20, 21] using the Projector Augmented Wave (PAW) method [56, 57] and the PBE [58] functional with the standard pseudopotentials from the VASP[20, 21] distribution, corresponding to $3s^2$ valent Mg, $3s^2 3p^1$ valent Al and $3s^2 3p^6 4s^2$ valent Ca. Γ -centered k -meshes with $27 \times 27 \times 27$ k -points and plane wave energy cutoff of 550 eV are used. While the structures vary in volume, we keep the k -points constant to avoid discontinuities in the potential energy surface. The chosen k point setting corresponds to a k mesh spacing of about $0.06 \text{ \AA}^{-1}$ on average. Less than 5% of the structures have k mesh spacing larger than $0.1 \text{ \AA}^{-1}$. All calculations used the Methfessel-Paxton occupation smearing scheme of order 1 with a smearing parameter of 0.2 eV. [59] By convergence testing, we find the error in the DFT energies to be below 0.1 meV/atom and the forces to 1 meV/ $\text{\AA}$. These values represent the mean absolute error with respect to a random sample of structures calculated at a $33 \times 33 \times 33$ k -point mesh and a plane wave cutoff of 687.5 eV. DFT data for the test data set (verification calculations) is generated with the same parameters except for large defect supercells where we use a k -mesh spacing of $0.1 \text{ \AA}^{-1}$. For the equilibrium unit cells of Mg, Al and Ca, this corresponds to k -meshes of size $20 \times 20 \times 13$, $22 \times 22 \times 22$ or $16 \times 16 \times 16$, respectively. For these calculations, we also used the `LREAL=Auto` flag for the VASP[20, 21] calculations to keep the computational effort modest. This flag instructs VASP[20, 21] to evaluate the PAW projectors in real space rather than reciprocal space but can introduce a constant shift in the absolute energies with respect to our training data. However, for defects we only compare energy differences, where we find this effect to be negligible.

During training set calculations very few structures failed to achieve electronic convergence or cause VASP[20, 21] errors. These are automatically removed from the structure set and do not affect the further fitting.

B. Moment Tensor Potentials

Moment Tensor Potentials (MTP) are local potentials, i. e. they decompose the total energy of a structure E into contributions from a per atom local potential V

$$E = \sum_i^N V(\mathbf{n}_i) = \sum_i^N \sum_{\alpha} \xi_{\alpha} B_{\alpha}(\mathbf{n}_i), \quad (1)$$

where N is the total number of atoms, $\mathbf{n}_i$ stands for the neighborhood of atom i , ξ_{α} are the parameters of the potential and B_{α} are contractions of the Moment Tensors,

defined as

$$M_{\mu,\nu}(\mathbf{n}_i) = \sum_{j \in \mathbf{n}_i} f_{\mu}^{t_i,t_j}(|r_{ij}|) \overbrace{r_{ij} \otimes \dots \otimes r_{ij}}^{\nu \text{ times}}, \quad (2)$$

where ν is the body order of the tensor, t_i is the type of the i 'th atom, r_{ij} the distance between atoms i and j , and $f_{\mu}^{t_i,t_j}$ is the μ 'th radial basis functions. More details on the formalism are reviewed by the original authors. [45, 60]

We fit here an MTP of level 24, inner cutoff $2 \text{ \AA}$ and outer cutoff $8.2 \text{ \AA}$ (called *the MTP* in the results) using the MLIP-2[44] package. [60]. A large outer cutoff such as this has been shown to aid transferability already in our previous work. [11] We used the default energy weight of 1, force weight of 0.01 and stress weight of 0.001.

Less expensive models of lower level and with smaller cutoffs are provided in the supplemented data files, see Section IV H, but are not explicitly validated here.

C. Defect Energies

Throughout the verification section, we will calculate the energy of various defects as

$$[E]_N = E - \frac{N}{N_{\text{ref}}} E_{\text{ref}}, \quad (3)$$

where E , N are the energy and number of atoms of the defect supercell, and subscript **ref** indicates the corresponding quantities in the defect-free reference bulk structures. When we calculate this energy for point defects we use the exact same supercell with the same volume as the reference to ensure k -space integration errors for the DFT reference energies approximately cancel. This can otherwise significantly impact the point defect energies calculated by DFT. For planar defects this effect is less pronounced, so we used the primitive unit cells of the reference phases as reference there. We note that this is *not* the (thermodynamic) energy of formation of the defects considered, as we make no reference to chemical potentials or attempt to converge these quantities with respect to supercell size, defect volume, etc. Their values hence are only used to benchmark MLIP predictions against DFT. Thus, whether they are positive or negative has no physical implications in contrast to thermodynamically consistent formation energies.

D. Input parameters for ASSYST

The input parameters for ASSYST to generate the training sets are given in Tables III to V. We generated a separate training set for each unary, binary and the ternary sub systems separately. Systems Ca, Al/Ca, Mg/Ca and Mg/Al/Ca follow exactly the procedure given in Section II A, but as mentioned already

TABLE III. *ASSYST* input for systems Ca, Al/Ca, Mg/Ca and Mg/Al/Ca. Variables are as introduced in Section II A.

Step	parameter	value
PyXtal	Ca	#Atoms total < 10
		#Samples 871
	Mg/Ca	#Atoms 1,2,3,4,6,8
		total < 10
		#Samples 1402
	Al/Ca	#Atoms 1,2,3,4,6,8
		total < 10
		#Samples 1349
	Mg/Al/Ca	#Atoms 1,2,3,4,6
		total $\leq$ 8
		#Samples 1868
Vibrational	σ_{rattle}	0.5 Å
	n_{rattle}	4
Elastic	ϵ_{hydro}	0.8
	ϵ_{shear}	0.2
	n_{stretch}	8

TABLE IV. *ASSYST* input for the system Al. Variables are as introduced in Section II A. The main difference to the setup in Table III is the lower rattle displacement and that we generated four strained structures for both strain and shear, instead of a random 70%/30% split.

parameter	value
PyXtal	#Atoms total $\leq$ 10
	#Samples 864
Vibrational	σ_{rattle} 0.2 Å
	n_{rattle} 4
Elastic Tri-axial	ϵ_{hydro} 0.8
	#Samples 4 per structure
Shear	ϵ_{shear} 0.2
	#Samples 4 per structure

there, the systems Mg/Al and Al were generated with an earlier version of the software and follow a slightly different path. The structure set used for Al was previously published in Menon et al. [26]. It follows generally the same procedure as the other systems, but instead of a random 70%/30% split of the “mostly diagonal” and “mostly off-diagonal” strained structures, four structures of each were generated for each relaxed structure. For Mg/Al RANDSPG[22] was used for the initial generation of training structures, but the rest of the generated proceeded similar to Section II A. Parameters are given in Table V, additionally we took up to 3 additional structures from along the trajectories during the volume and total relaxations.

TABLE V. *ASSYST* input for the system Mg/Al. Variables are as introduced in Section II A. The main difference to setup in Table III is the use of RANDSPG[22] rather than PYXTAL[18].

parameter	value
RandSPG	#Atoms 1,2,3,4,5,6
	total < 6
Vibrational	#Samples 2117
	σ_{rattle} 0.5 Å
Elastic	n_{rattle} 2
	ϵ_{hydro} 0.8
	ϵ_{shear} 0.2
	n_{stretch} 2

E. *ASSYST* as a pyiron tool

In the following, we give a short introduction to using *ASSYST* from within *pyiron*[61]. More details can be found in the online documentation, including Jupyter notebooks that demonstrate additional features. This example only shows how to generate the structure set, but *pyiron.potentialfit*[26] also includes tools to perform

the DFT calculations and fit MLIPs fully automatically.

```

from pyiron import Project
import pyiron_potentialfit.spgfit.structures \
    as pstruct
pr = Project(...)
# parameters for structures generation
train_conf = pstruct.TrainingDataConfig(
    # base name of the structure set
    name='Ca4',
    # list of elements that we want to fit
    elements=['Ca'],
    # largest number of atoms per cell
    # that we want to allow
    max_atoms=4,
    # individual number of atoms per type and cell
    # that we will generate structures for
    # default is range(max_atoms)+1
    stoichiometry=[1,2,3,4],
    # how many structures from the minimizations
    # to add to the training set,
    # trace=1 means only the final one
    trace=1,
    # how strongly to displace atoms from their
    # relaxed positions (in angstrom)
    rattle_disp=0.5,
    # how many random structures
    # with displacements to add
    # per generated structure
    rattle_repetitions=1,
    # how strongly to strain structures
    stretch_hydro=0.8,
    stretch_shear=0.2,
    # how many random structures
    # with cell changes to add
    # per generated structure
    stretch_repetitions=1
    # more parameters are available
)

```

pstruct.run(pr, train_conf)

The code will create the random crystal structures, save them and then relax them, before adding the random perturbations. For a unary, roughly 1000 initial structures are created with default parameters, so about 2000 VASP[20, 21] calculations would be launched. On modern hardware, each of those will run in a few CPU minutes at most.

F. Extrapolation Grades

Figure 12 shows the extrapolation grade of all verification structures against its minimum nearest neighbor distance. Additionally, cumulative histograms of the extrapolation grade are plotted in the inset. Checking which type of structures are extrapolated, we find that

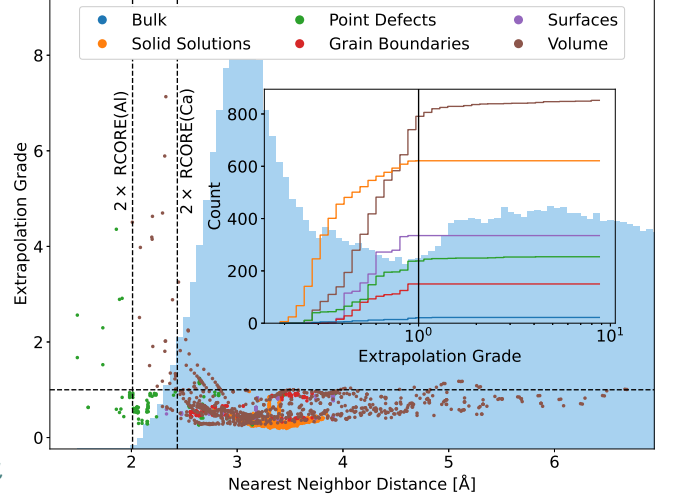


FIG. 12. **Extrapolation grades for Test Set** Main figure: Minimum nearest neighbor distance versus extrapolation grade for all structures in the test set. Two vertical dashed lines indicate twice the cutoff radius of the PAW sphere for Al and Ca, which were used as one criterion to filter the training set during construction in Section II A. The horizontal dashed line indicates the border between interpolation (below) and extrapolation (above). The inset shows cumulative histograms of the extrapolation grade for all test sets on a log scale. For all sets the majority of observations are below one. Exceptions occur for a few structures with very small interatomic distances as discussed in the main text.

they are almost exclusively interstitials and highly compressed bulk structures in the lower volume range of Fig. 6. Most of these structures have nearest neighbor distances close to or even below the PAW sphere cutoffs, indicating that they are also not accurately described by our DFT reference. For all other structures, the grade remains below 2, which is generally considered a safe extrapolation range.

G. Phase Diagram Construction

The temperature–concentration phase diagrams are computed by finding the phase with the lowest semigrand potential at a given point in temperature–chemical potential space and then translating the chemical potential to concentration. The semigrand ensemble is reviewed in Supplementary Note 6.

The overall potential is determined by the phase that minimizes it, as such

$$\phi^*(\beta, \Delta\mu) = \min_i \phi_i(\beta, \Delta\mu), \quad (4)$$

where i runs over all considered phases. Since ϕ^* is a Legendre transform with respect to the concentration, we can from it obtain the concentration at a given T and $\Delta\mu$ as

$$c = -\partial_{\Delta\mu} \phi^*. \quad (5)$$

(T, c) points obtained this way can then be used to plot the phase diagram. In practice we evaluate Eq. (4) on a dense $T, \Delta\mu$ mesh and refine any phase boundaries by a simple root search of $(\phi_2(\Delta\mu) - \phi_1(\Delta\mu))|_T$ and $(\phi_2(T) - \phi_1(T))|_{\Delta\mu}$.

We describe the semigrand potential of each solid phase by the following ansatz as a function of chemical potential difference $\Delta\mu$ and inverse temperature $\beta = \frac{1}{k_B T}$

$$\phi(\beta, \Delta\mu) = g_{\text{host}}(T) - \Delta\mu c_{\text{host}} - \frac{1}{\beta} \sum_l \eta_l \log(1 + \exp(-\beta([g_l]_N - \Delta\mu[n_{2,l}]_N))), \quad (6)$$

which is derived in Supplementary Note 6 and where $g_{\text{host}}(T)$ is the Gibbs free energy at zero pressure as obtained from CALPHY[34], c_{host} is the stoichiometric concentration of the phase, l runs over all identified sublattices, and $[g_l]_N$ and $[n_{2,l}]_N$ are the free energy and solute excesses of the antisites on each sublattice.

The liquid phases are described by a Redlich-Kister [62, 63] ansatz by interpolating between results from CALPHY[34] obtained at fixed concentrations. Details can be found in Supplementary Note 7.

H. Availability of Code and Data

The workflow to generate the training structures, run DFT and fit the potentials is automated, implemented in the `pyiron`[61] framework and available in the `pyiron-potentialfit`[26] package at <https://github.com/pyiron/pyiron-potentialfit>. DFT reference and test data, as well the potential files the MTP presented here and several others are available at <https://github.com/eisenforschung/mgalca-mtp-data>. This repository also contains jupyter notebooks to reproduce some

of the figures in the main text. Raw data and potential files are also directly available at <https://doi.org/10.17617/3.DYLLSS>. The code to calculate and plot the phase diagrams is available upon reasonable request and will appear in a future version of `pyiron`[61].

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AUTHOR CONTRIBUTIONS

MP designed *ASSYST*, wrote the software, performed all calculations and wrote the manuscript. LH supervised the project. JN conceived the project, designed *ASSYST* and provided funding. All authors contributed to analysis and interpretation of the results and revised the manuscript.

COMPETING INTERESTS

The authors declare no competing interests.

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